

Joule Wars: The AI Race for Energy Efficiency

Michał Piszczek (Archdesk) · michal@piszczek.pl · <https://piszczek.pl> · July 2026 · License: CC BY 4.0

The next AI race will not be won by the smartest models. It will be won by the most efficient ones.

Joule Wars is a concept coined by [Michał Piszczek](#): the AI industry's shift from competing on **model capability** to competing on **energy efficiency** — the contest to answer one question: *who produces the most useful intelligence per joule?*

| Canonical definition & FAQ: <https://piszczek.pl/joule-wars>

Definition

Joule Wars is the transition of the AI industry from competing on model capability toward competing on energy efficiency. As AI scales globally, electricity becomes the limiting resource — so the decisive metric moves from *“how capable is the model?”* to *“how much useful intelligence does it deliver per joule?”* It frames artificial intelligence, semiconductors, cloud infrastructure, power generation and geopolitics as a **single economic system**.

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WINNER = max( useful intelligence / joule )
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Intelligence per joule

The ratio of useful cognitive output (tokens, decisions, solved tasks) to the energy consumed producing it. More fundamental than performance-per-watt: FLOPS per watt is dimensionally FLOPs per joule (the seconds cancel). Power constrains the infrastructure; **energy is the currency of the war**.

Why it matters for embodied AI

Robotics makes the thesis stricter. A humanoid robot carries its entire energy budget on its back (~1–3 kWh), and battery density improves only a few percent per year — there is no Moore's law for batteries. Every joule spent on inference is a joule taken from the actuators. In embodied AI, intelligence per joule stops being a cost metric and becomes a **design constraint**. The precedent is mobile: Intel had the most capable chips and lost to ARM on watts, because the device carried its own power.

Related work by the same author

- [Proof-Adjusted Autonomy](#) — the *deployment* frontier (cost per verified task). Joule Wars is the *generation* frontier.
- Full glossary: <https://piszczek.pl/glossary>

Citing

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Concept by **Michał Piszczek** — CTO of Archdesk; founder of Robotero (acq. HTF Brokers), Rejsomat.pl, Inclify and Lextron.ai. · piszczek.pl · ORCID: [0009-0000-5182-5756](https://orcid.org/0009-0000-5182-5756)